Question	Answer	Marks
2(a)	no intermolecular forces (so no potential energy)	**B1**
2(b)(i)	mean square speed (of molecule(s))	**B1**
2(b)(ii)	kelvin/thermodynamic/absolute <u>temperature</u>	**B1**
2(c)(i)1.	$pV = NkT$	**C1**
	$4.7 \times 10^{-2} \times 2.6 \times 10^{5} = N \times 1.38 \times 10^{-23} \times 446$	**C1**
	or	
	$pV = nRT$ **and** $N = nN_A$ $4.7 \times 10^{-2} \times 2.6 \times 10^{5} = n \times 8.31 \times 446$ $n = 3.3$ (mol)	**(C1)**
	$N = 3.3 \times 6.02 \times 10^{23}$	**(C1)**
	$N = 2.0 \times 10^{24}$	**A1**
2(c)(i)2.	average increase $= 2900 / (2.0 \times 10^{24})$ $= 1.5 \times 10^{-21}$ J	**A1**
2(c)(ii)	$\Delta E_K = (3/2)k\,(\Delta)T$ $1.5 \times 10^{-21} = (3/2) \times 1.38 \times 10^{-23} \times (\Delta)T$	**C1**
	$(\Delta)T$ in range 70–72 K	**C1**
	$T = 173 + 273 + 70$ $= 520$ K	**A1**

Question	Answer	Marks
3(a)	(during melting,) bonds between atoms/molecules are broken	B1